

PROJECT DOCUMENTATION

Animal Diet & Nutrition Tracking
WaveWare
April 21, 2026

TEXAS STATE AQUARIUM

PUBLIC PORTFOLIO EDITION - OCTOBER 29, 2026 | NOT OFFICIAL AQUARIUM DOCUMENTATION

Public portfolio edition

Source and scope

Adapted with the project owner’s approval from the April 21, 2026 WaveWare Project Documentation. This public edition was prepared October 29, 2026. It documents an academic/portfolio Microsoft Access project based on Texas State Aquarium requirements. It is not official aquarium documentation, the actual final submission, a current aquarium system, or animal-care guidance.

Documented features are not verified outcomes

Workflow steps and feature descriptions record the source guide’s intended design. They do not prove that the downloadable near-final demo implements every function correctly. No native Microsoft Access retest, calculation validation, security assessment, production deployment or measured efficiency result is claimed.

Privacy and screenshot labels

The employee first and last name shown in a screenshot were permanently replaced with the generic labels Demo and Employee. The identifier and role remain visible. These labels do not establish a name-to-record mapping in the separate release database. Other project records, including animal names, dates and numeric values, remain in screenshots; this is not a fully anonymized or synthetic document.

Version and safety limits

Screenshots describe the source project version and may differ from the repository’s near-final release. Use a working copy of the download in Microsoft Access on Windows. Review macros and code before enabling active content. Demo passwords are intentionally public and provide no confidentiality. Do not use this guide for animal diets or care decisions.

Editorial changes

The original cover date is retained. Contents page numbers, clickable navigation and page labels were rebuilt; overview/design/conclusion wording was qualified for public sharing. The original PDF and the existing Access database release were not modified.

Contents: project overview

Page numbers refer to this public PDF edition. Select a row to jump to the section. Screens and workflows may differ from the released near-final demo.

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Contents: core operations

Page numbers refer to this public PDF edition. Select a row to jump to the section. Screens and workflows may differ from the released near-final demo.

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Executive summary

Project context

WaveWare is a Microsoft Access 365 project for organizing animal-diet and nutrition-related records around Texas State Aquarium requirements. The source report describes a workflow involving spreadsheets, manual calculations and handwritten records as the motivation for a proposed centralized application.

Documented approach

The proposed design brings together animal and employee records, food lots, diet entries, weight logs, inventory retrieval and reporting. The source guide describes menu-driven forms, nutritional calculations, diet-change forms and alert screens.

Portfolio value

The project illustrates relational database design, operational workflow analysis, form-based navigation and user documentation. The screenshots and instructions provide a way to review the design without opening Access.

Evidence limits

The repository download is a near-final representative demo, not the actual final submission. The guide does not establish current aquarium use, achieved accuracy or efficiency gains, secure access control, scalability, or clinical suitability. Native Access workflows and calculations have not been re-tested after the public-sharing edit.

Introduction

Purpose

The source project explores how a relational database could organize the information needed for diet planning, lot inventory, weight recording and reporting in one Access application.

Intended scope

The documented modules cover animal management, employee management, food and lot records, diet planning, diet adjustments, weight entry, alerts and reports. These are design and user-guide descriptions, not independent verification of every released feature.

How to read this guide

The design section explains the proposed organization of records and screens. The user manual retains source-version screenshots and navigation steps. Refer to the repository README for the public download, software requirements, demo passwords and limitations.

Version boundary

The screenshots originate from the April 2026 documentation. The separately published near-final database may differ. No change to that database is made by publishing this guide, and no animal-care or nutrition recommendation is provided.

System analysis & design

Design overview

The source report describes a centralized, relational approach to records that would otherwise be spread across manual processes. The proposed workflow follows food lot receipt, animal records, diet composition, diet changes, inventory use and weight review.

1. Food lot entry

The intended process records lot identification, received quantity and nutritional inputs so lots can be referenced in diet planning and inventory tracking.

2. Animal management

Animal records hold identifiers, species and weight-related fields used by the proposed diet and monitoring workflow.

3. Diet creation

The documented forms select animals, food items, lots and quantities. Calorie and nutrition calculations are described as intended application behavior; their accuracy is not verified here.

4. Diet adjustment

The source guide describes forms for changes associated with food lots and animal weights, including transitions between lots. These steps are not animal-care instructions.

5. Inventory tracking

Lot retrieval records are intended to connect food usage with available quantities. Inventory-update behavior requires native Access validation.

Workflow & database design

6. Weight monitoring and alerts

The documented workflow records weights and compares them with stored ranges. The guide describes alert screens, but does not verify alert timing, thresholds or clinical appropriateness.

7. Reporting

The source describes reports covering animals, diets, weights, food and inventory. Report outputs and query calculations were not re-tested for this publication.

Conceptual record groups

Animal: identifiers, names, species and weight-related values. Employee: identifiers and roles associated with data entry. Food: food items and animal suitability. Lot: received food lots, nutrition inputs and quantities.

Diet and inventory history

Diet: daily entries, selected foods, lots and quantities. Diet Change: proposed changes over time. Lot Retrieval: quantities taken from lots. Animal Weight Log: historical weight entries. These are conceptual labels from the report, not a verified full schema or exact object-name inventory.

Design intent

Relationships and structured records are intended to improve consistency and reduce duplication. This document does not establish that every constraint, calculation or transaction behaves correctly in the public demo.

Relationships & interface

Relationships described in the source

The source describes one animal having multiple diet records and weight logs, and one food lot being referenced by multiple diet entries and retrieval records. Employee references are intended to associate records with the person entering them.

Integrity limits

These relationships explain the design approach. The guide is not a replacement for inspecting the Access relationships, field types, indexes, validation rules and saved queries. It does not independently verify referential integrity or transaction behavior.

User interface

A main dashboard and separate animal, employee, lot and report menus organize navigation. Forms group related fields, while search controls and record-navigation buttons support browsing and editing.

Reviewing usability

The source presents a consistent interface as a design goal. No usability study, accessibility audit, measured time savings or user acceptance result is claimed. The following screenshots show the documented interface, not proof of production use.

Automation & considerations

Documented automation

The source guide describes calorie calculations, inventory updates following lot retrieval, weight-range alerts and structured diet-change forms. Treat these as intended or documented features until verified in the relevant Access version.

Design goals

The project aims for structured input, reduced duplication, consistent navigation and room for additional records. These are goals rather than measured accuracy, performance, reliability or scalability results.

Validation needed

Before relying on any workflow, compare the guide with the downloaded demo; inspect saved queries, macros and formulas; check units and representative inputs; and verify resulting records and reports in Microsoft Access on Windows.

Public-demo boundary

The release retains project records and intentionally public demo passwords. It is not a secure production deployment, a fully anonymized dataset or animal-care guidance. The current publication adds documentation only and does not modify or re-test the database.

USER MANUAL

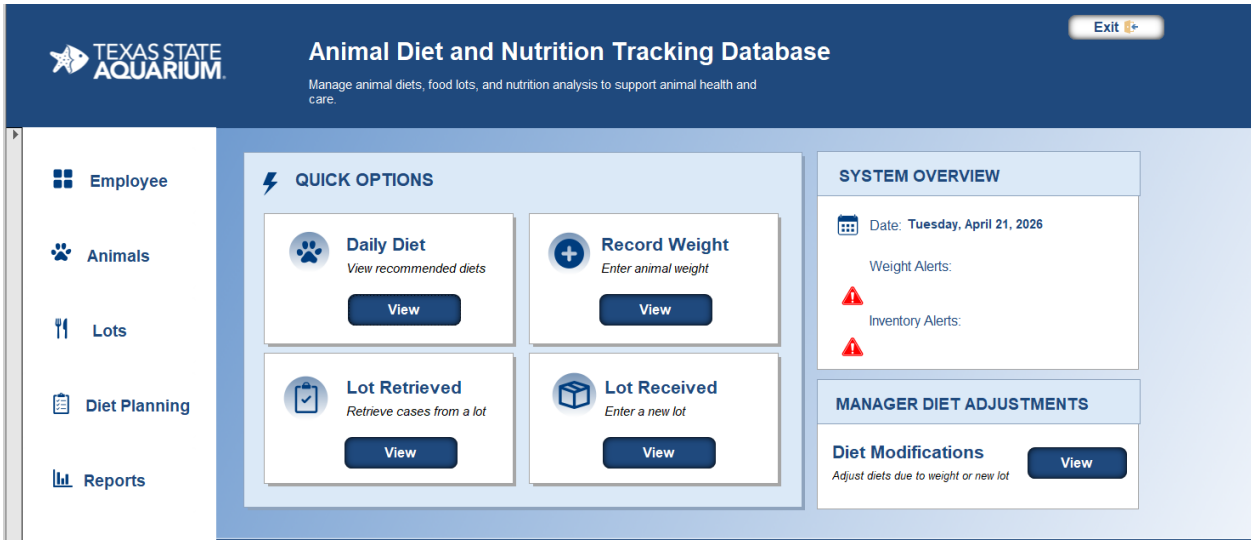
Overview

This guide describes the proposed Access interface for animal records, food lots, diets, weights and reports. It is a navigation reference, not a record of production deployment or verified behavior in the released demo.

The system is organized into several main functional areas accessible through the dashboard:

- Animal Management
- Employee Management
- Lot (Inventory) Management
- Diet Planning
- Reporting
- Alerts

1. MAIN DASHBOARD (frmMain)



Purpose: The dashboard serves as the central navigation hub for the entire system.

Features:

- Quick access buttons:
  - Daily Diet
  - Record Weight

  - Lot Retrieved
  - Lot Received
- System Overview:
  - Displays current date
  - Shows alerts (weight & inventory)
- Manager Diet Adjustments:
  - Access to diet change features

How to Use:

1. Open the database → the Main Dashboard will appear.
2. Select a function using the buttons:
   a. Click **View** under Daily Diet to access diet planning.
   b. Click **View** under Record Weight to log animal weights.
   c. Click **View** under Lot Received to add new food lots.
   d. Click **View** under Lot Retrieved to remove inventory.

2. ANIMAL MANAGEMENT

a. Animal Menu (frmAnimalMenu)



Purpose: Provides access to animal-related functions.

Options:

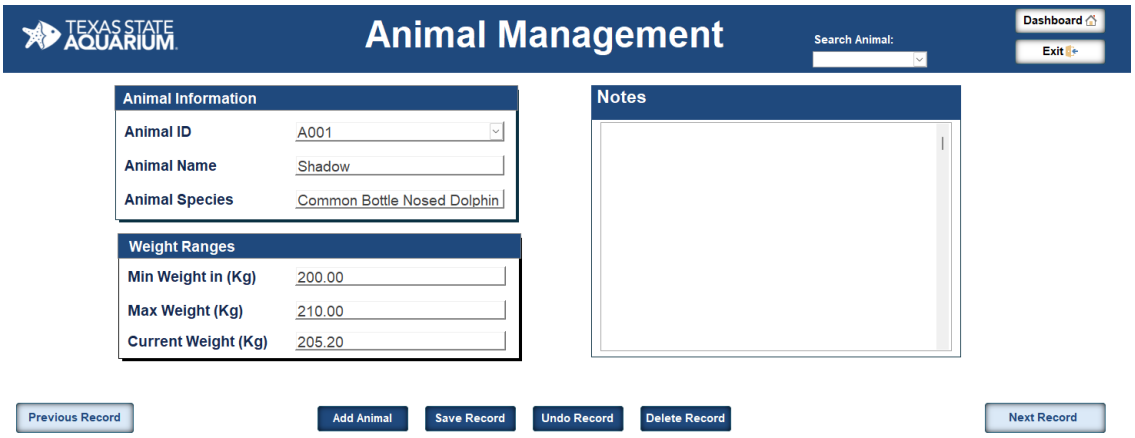
- View Animal Records
- Record Animal Weight

- Add New Animal

Steps:

1. Click **Animals** on the dashboard.
2. Select desired option.

b. Animal Form (frmAnimal)



Purpose: Add, edit, or view animal records.

Fields: Animal ID, Animal Name, Animal Species, Min Weight, Max Weight, Current Weight, Notes

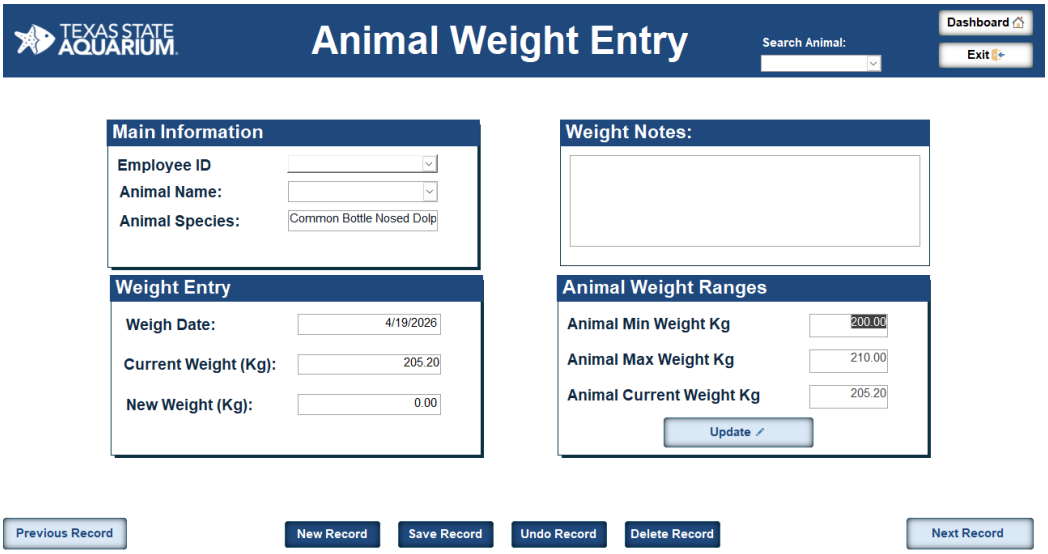
Steps:

1. Click **Add Animal** to create a new record.
2. Enter all required information.
3. Click **Save Record**.

Navigation:

- Use **Previous Record / Next Record** to browse animals.
- Use **Search Animal** dropdown to quickly locate a record.

3. ANIMAL WEIGHT ENTRY (frmAnimalWeight)



Purpose: Record and update animal weights.

Sections: Employee & Animal Info, Weight Entry, Weight Ranges

Steps:

1. Select **Animal Name** from dropdown.
2. Enter:
    a. Weight Date
    b. New Weight
3. Click **Update** to update current weight.
4. Click **Save Record**.

Notes:

- System compares weight against min/max values.
- Out-of-range weights trigger alerts.

4. EMPLOYEE MANAGEMENT

a) Employee Menu (frmEmployeeMenu)



Options:

- View Employee Records
- Add New Employee

b) Employee Form (frmEmployee)



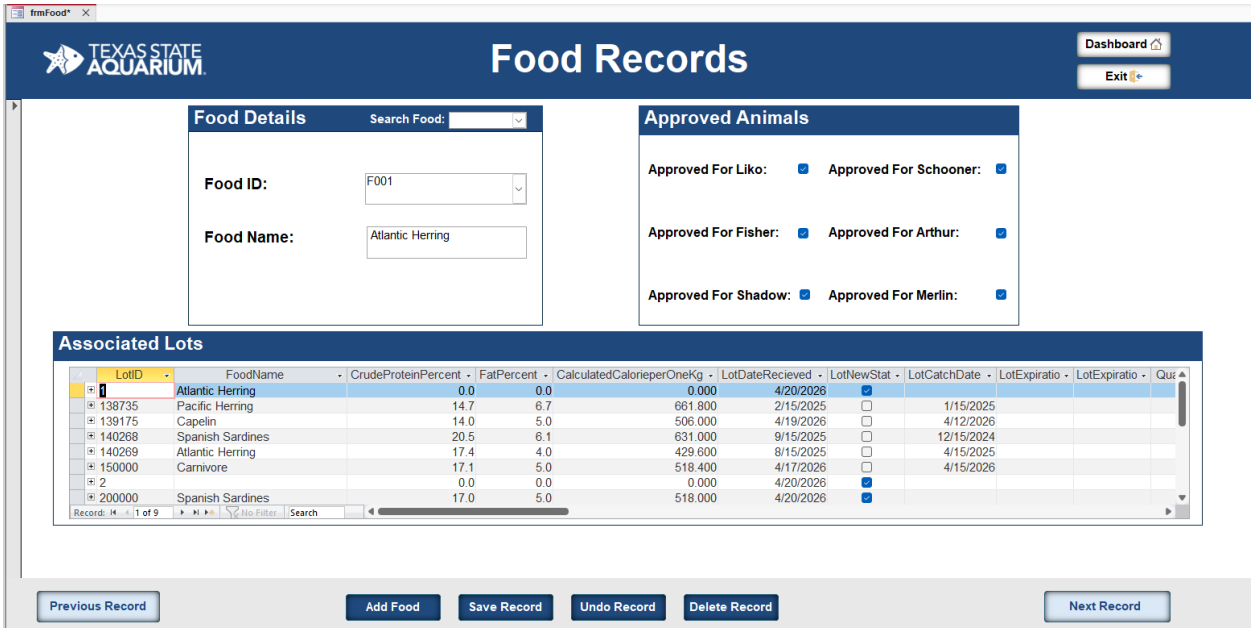
Purpose: Manage employee records.

Fields: Employee ID, First Name, Last Name, Role

Steps:

1. Click **Add Employee**.
2. Enter details.
3. Click **Save Record**.

5. FOOD MANAGEMENT (frmFood)



Purpose: Manage food items and assign them to animals.

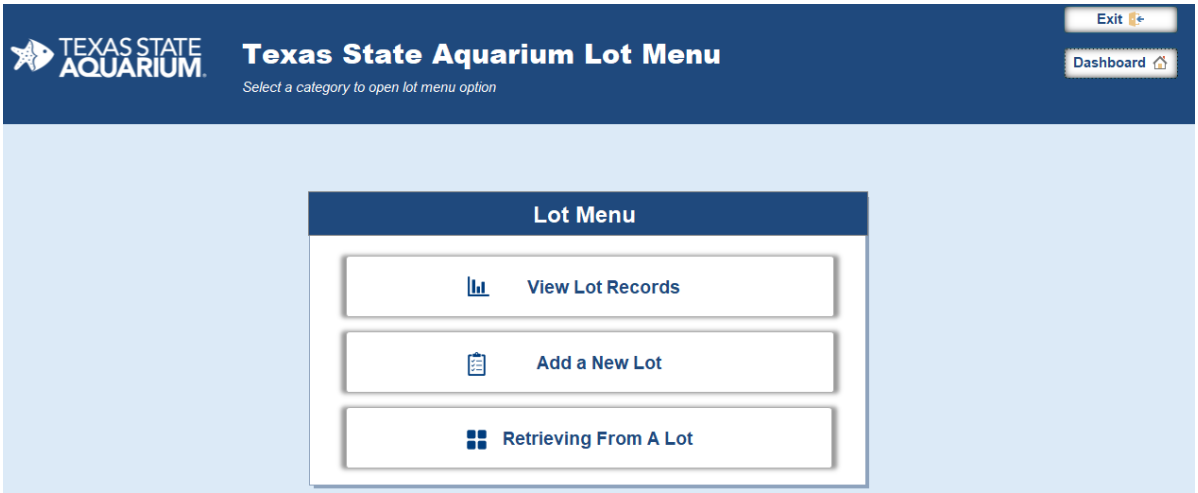
Features: Food ID & Name, Approved Animals (checkboxes), Associated Lots (subform)

Steps:

1. Click **Add Food** to create new food.
2. Enter food name.
3. Select animals that can consume the food.
4. Click **Save Record**.

6. LOT MANAGEMENT

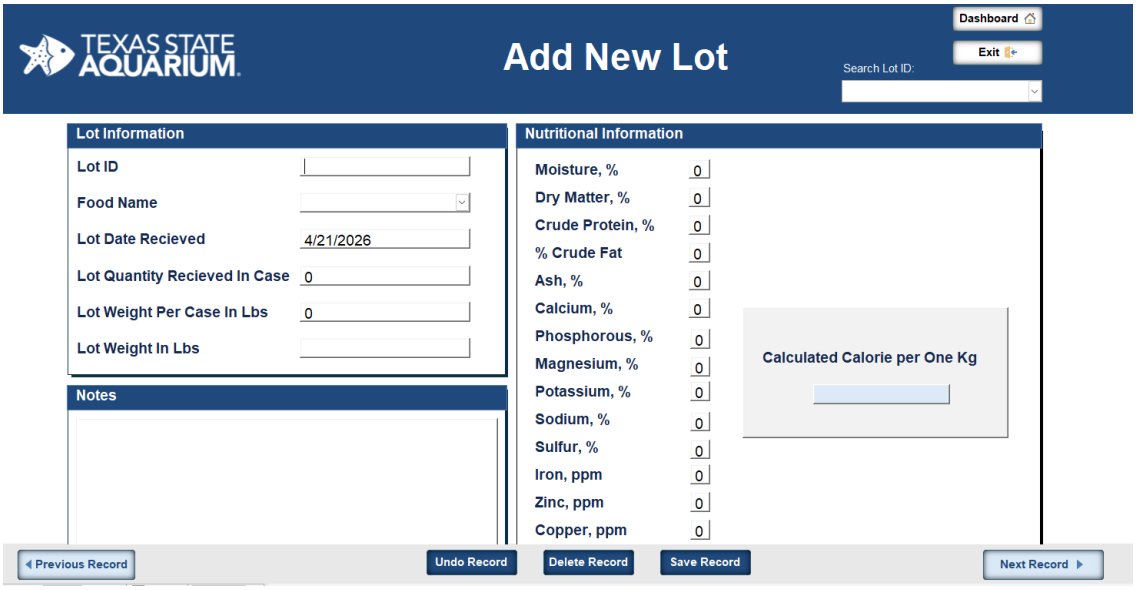
a) Lot Menu (frmLotMenu)



Options:

- View Lot Records
- Add New Lot
- Retrieve From a Lot

b) Add New Lot (frmAddNewLot)



Purpose: Enter new food inventory.

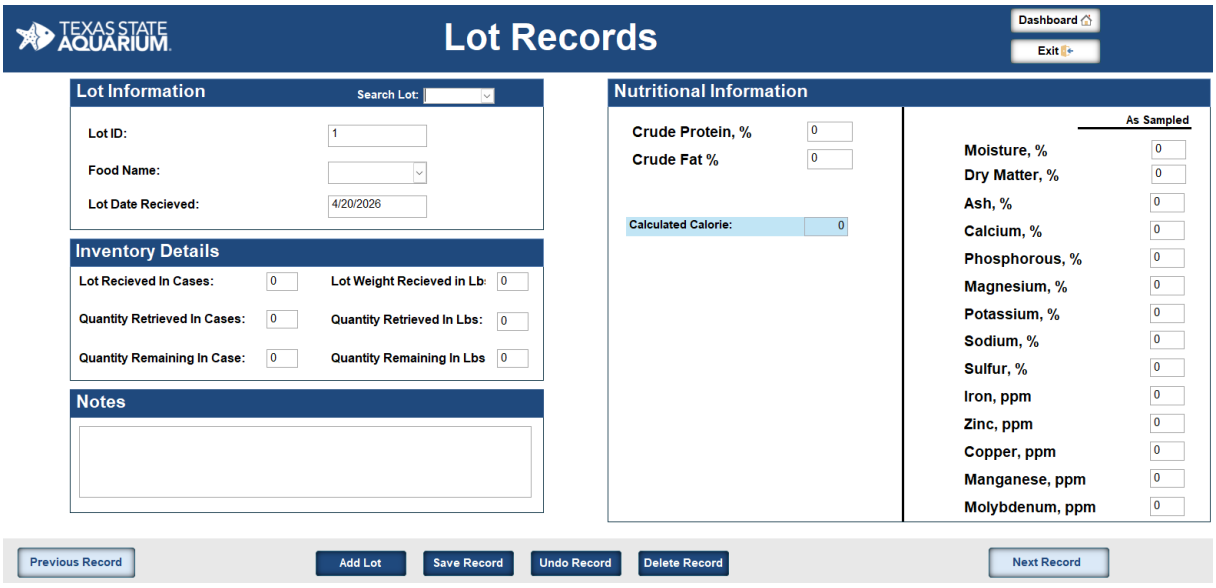
Sections: Lot Information, Nutritional Information, Notes

Steps:

1. Enter:
    a. Lot ID
    b. Food Name
    c. Date Received
    d. Quantity (cases)
    e. Weight per case
2. Enter nutritional values.
3. Click **Save Record**.

Notes: Calories per kg are automatically calculated.

c) 6.3 Lot Records (frmLotRecords)



Purpose: View and manage existing lots.

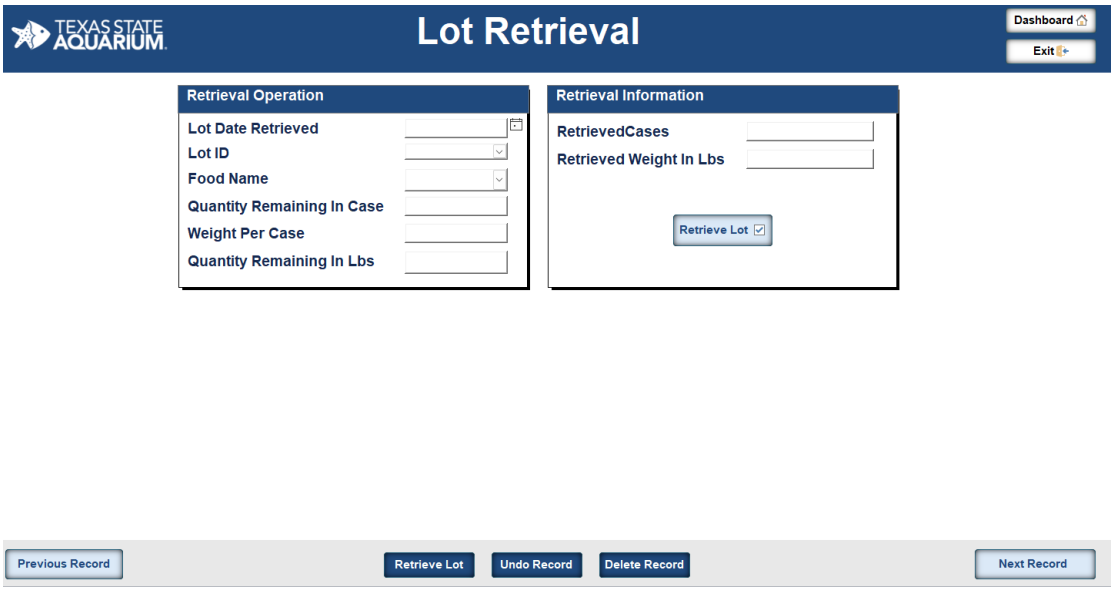
Features: Inventory tracking:

- Quantity Received
- Quantity Retrieved
- Remaining inventory

Steps:

1. Use search dropdown to locate a lot.
2. View or update details.
3. Click **Save Record** if changes are made.

d) 6.4 Lot Retrieval (frmLotRetrieval)



Purpose: Remove food from inventory.

Steps:

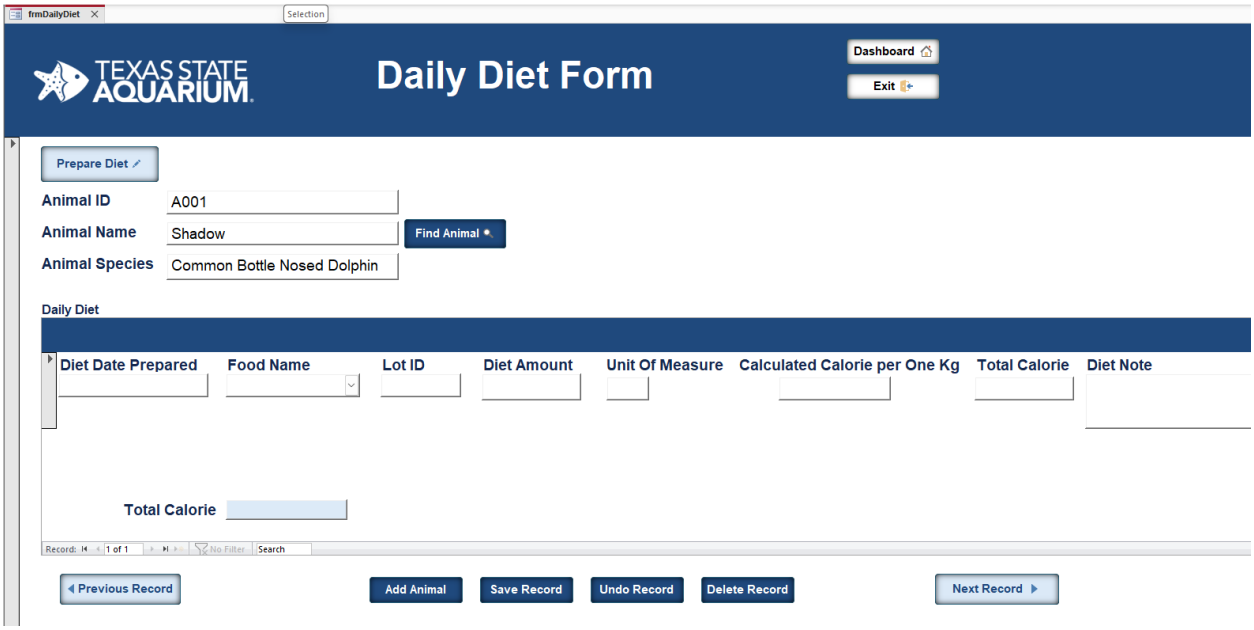
1. Select:
    a. Lot ID
    b. Food Name
2. Enter:

a. Retrieved Cases
b. Retrieved Weight

3. Click **Retrieve Lot**.
4. Click **Save Record**.

7. DIET PLANNING

a) Daily Diet Form (frmDailyDiet)



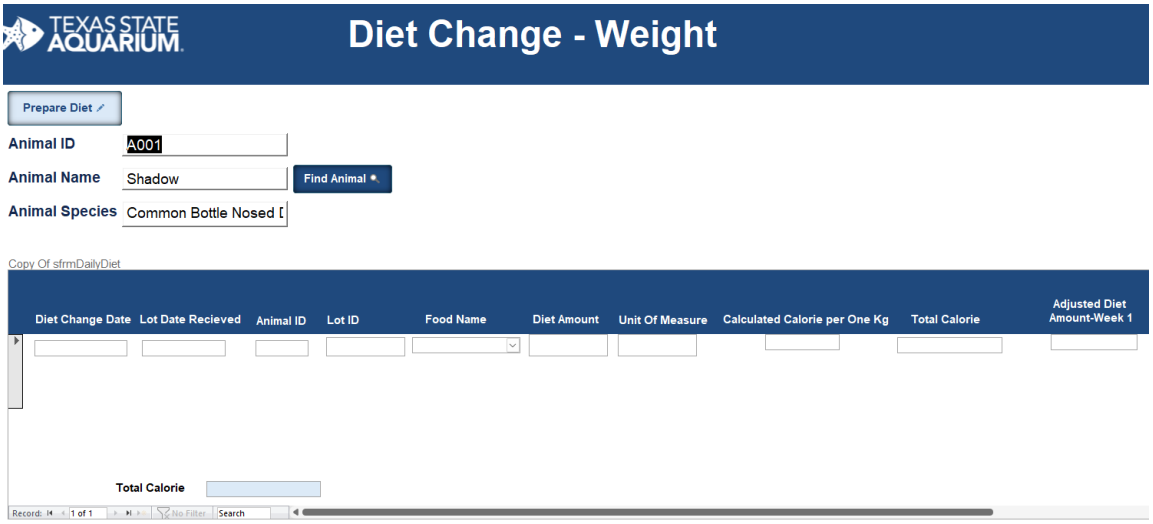
Purpose: Create and manage daily diets for animals.

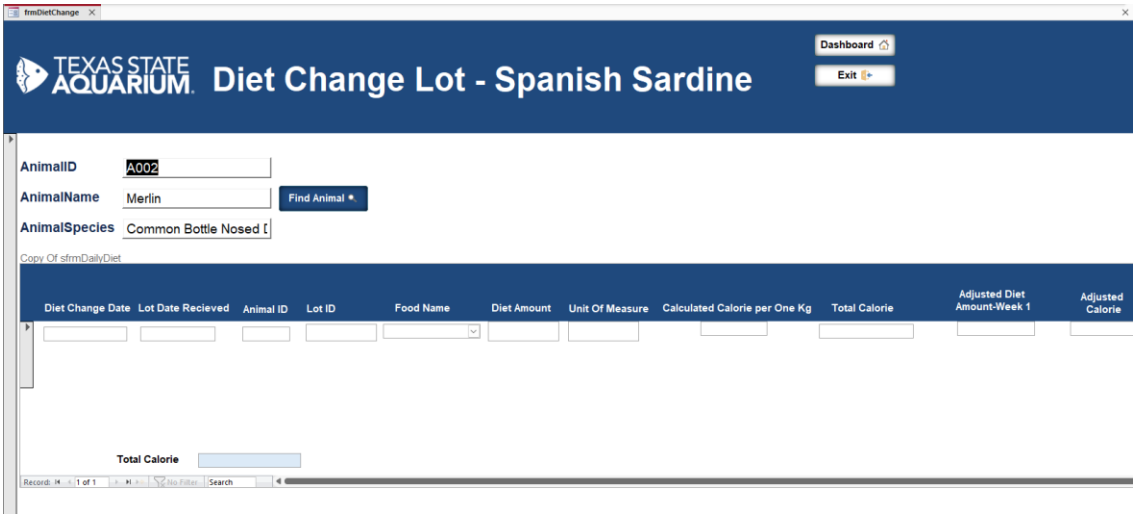
Fields: Animal ID, Food Name, Lot ID, Diet Amount, Unit of Measure, Calories

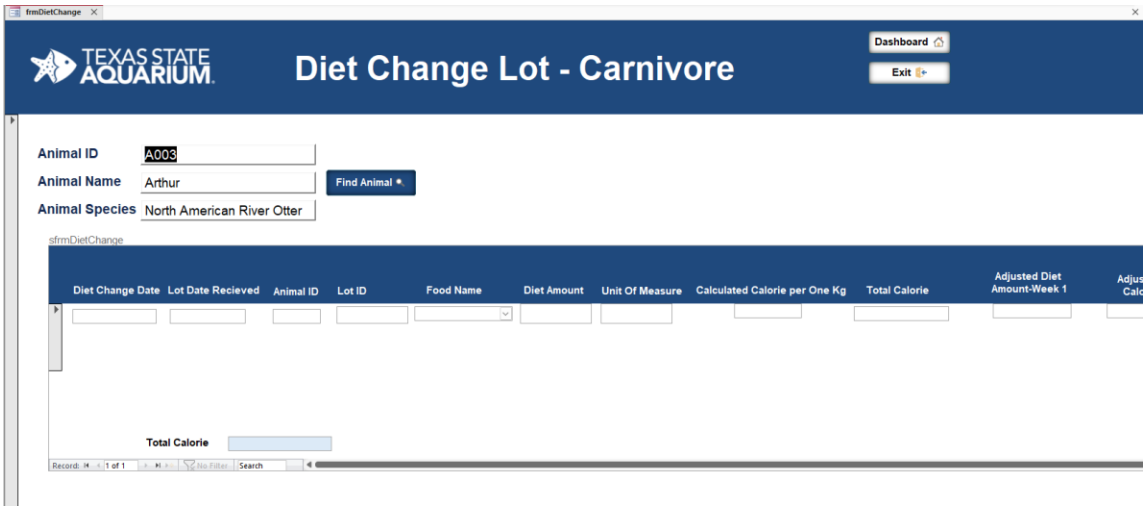
Steps:

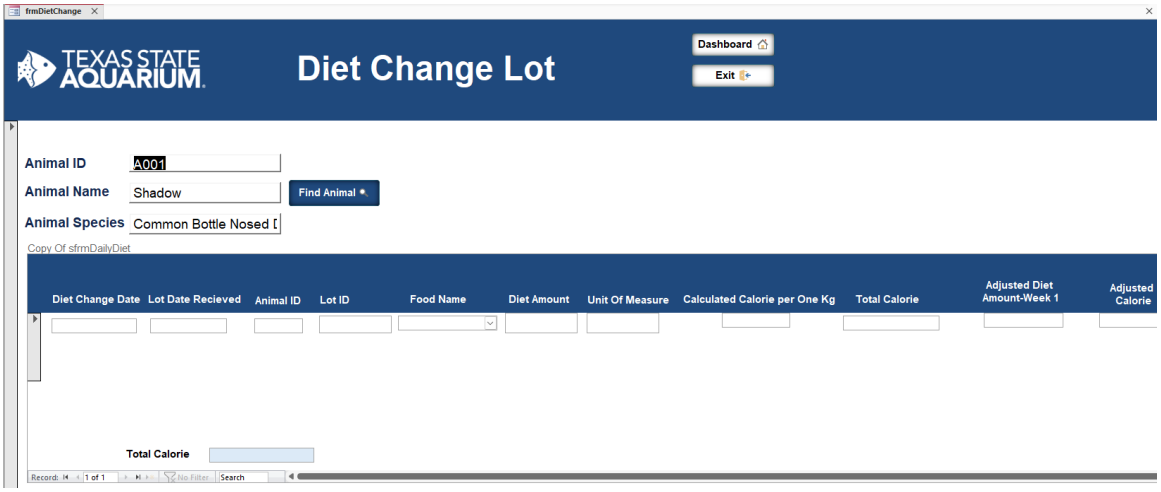
1. Click **Prepare Diet**.
2. Select animal.
3. Add food items and quantities.
4. System calculates total calories.
5. Click **Save Record**.

b) Diet Change Forms (frmDietChange + variations)









Purpose: Adjust diets based on:

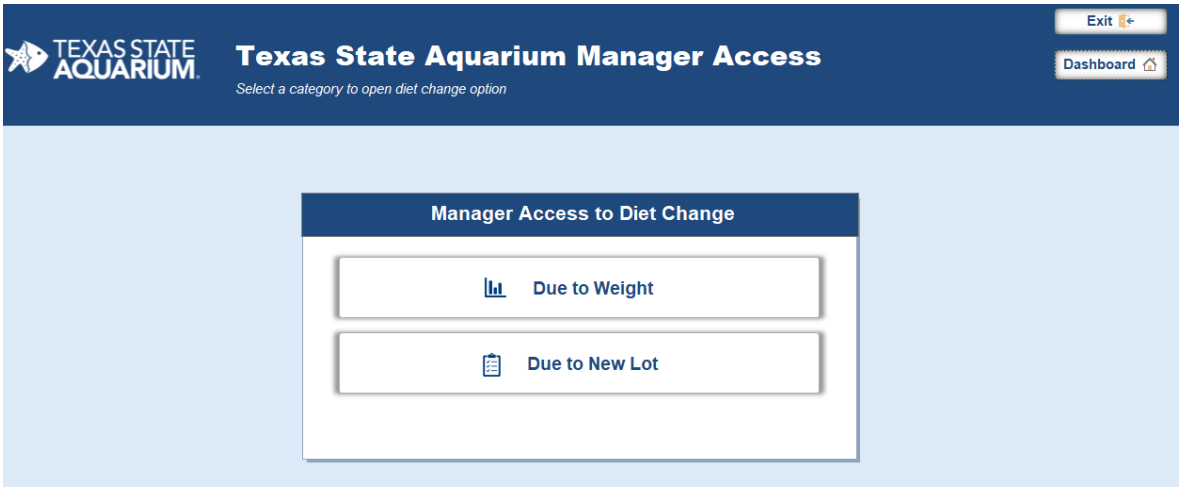
1) New food lots
2) Animal weight changes

Types: General Diet Change, Carnivore Diet Change, Spanish Sardine Diet Change, Weight-Based Diet Change

Steps:

1. Select animal.
2. Review current diet.
3. Adjust:
    a. Food amounts
    b. Percentages
4. Save updated diet.

c) Manager Diet Adjustments (frmDietMenuManagers)

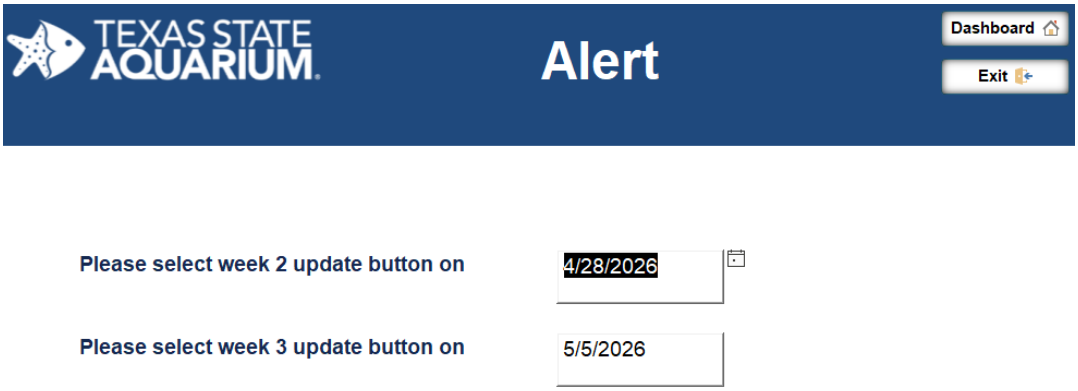


Purpose: Provide controlled access for diet modifications.

Options:

- Adjust Diet Due to Weight
- Adjust Diet Due to New Lot

8. ALERT SYSTEM (frmAlert)



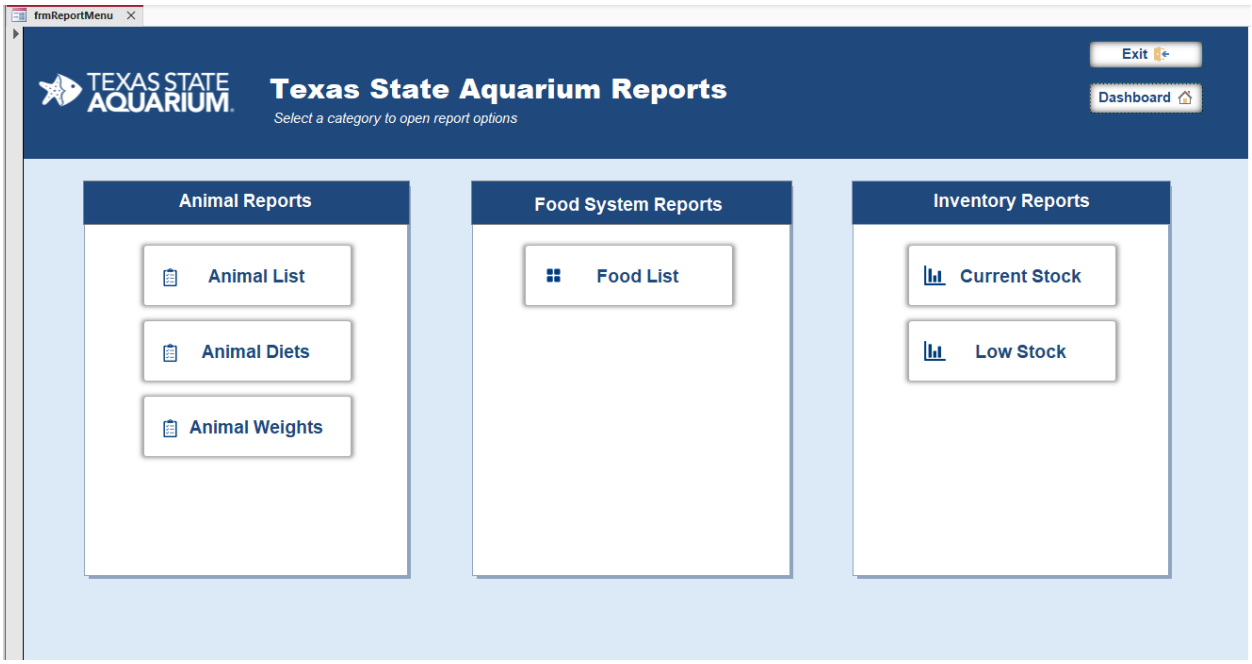
Purpose: Notify users of required updates.

Alerts: Weight alerts, Diet adjustment alerts

Steps:

1. Open Alerts screen.
2. Review dates for required updates.
3. Perform necessary diet changes.

9. REPORTS (frmReportMenu)



Purpose: Generate system reports.

Categories: Animal Reports, Food Reports, Inventory Reports

Available Reports:

- Animal List
- Animal Diets
- Animal Weights
- Food List
- Current Stock
- Low Stock

Steps:

1. Open Reports menu.
2. Select report type.
3. View or print report.

10. GENERAL NAVIGATION FEATURES

- **Dashboard Button:** Returns to main screen
- **Exit Button:** Closes current form
- **Previous/Next Record:** Navigate records
- **Search Bars:** Quickly locate records
- **Undo Button:** Revert changes
- **Delete Button:** Remove records

These are documented workflows and intended benefits. Native Access testing, calculation accuracy, reporting behavior and operational outcomes have not been verified for the public demo.

Conclusion & next steps

What the project documents

WaveWare presents a relational Access design for connecting animal records, food lots, diet entries, weight logs, inventory activity and reporting. The guide makes the documented screens and intended workflow easier to review.

What is not established

Publication does not prove that the system is used by the aquarium, meets all requirements, improves operational results, calculates nutrition correctly, or is production-ready. The near-final representative demo is separate from the April 2026 source documentation.

Recommended follow-up checks

In a working copy on Windows, compare the documented forms with the release, inspect queries and macros, test representative diet and retrieval entries, verify units and totals, and compare reports with underlying records. These are proposed checks, not tests completed during publication.

Safe use

Preserve a clean copy of the download and review active content before enabling it. Public demo passwords are not security controls. Do not use the database, screenshots or documentation as animal-care or feeding guidance.